Section D — AI Interaction Log & Reflection

D1. Interaction Log

#	Tool Used	Prompt / Query	Output Received	What I Accepted / Modified / Rejected — and Why
1	ChatGPT (GPT-5.5)	Suggest suitable models for chargeback prediction.	Recommended MLP, Random Forest, and XGBoost.	Accepted MLP recommendation but compared it with other models before making a final decision.
2	Claude Sonnet	Explain class imbalance in fraud detection datasets.	Discussed recall, precision, and minority classes.	Accepted explanation and used it to justify focusing on recall instead of accuracy.
3	Gemini	Review provided fraud detection code for issues.	Identified possible data leakage and imbalance concerns.	Accepted the data leakage observation after verifying it against lecture materials.
4	ChatGPT (GPT-5.5)	Suggest techniques for improving fraud detection recall.	Recommended SMOTE, class weights, and threshold tuning.	These techniques applied and compared their impact on model performance.
5	Claude Sonnet	Interpret confusion matrix results.	Explained false positives and false negatives.	Modified explanation to reflect MyMart's RM 85 chargeback cost.
6	Gemini	Suggest ethical risks in fraud detection systems.	Mentioned bias, fairness, and customer impact.	Accepted the ideas but linked them specifically to BNPL users and new customers.
7	ChatGPT (GPT-5.5)	Improve report wording and executive summary.	Produced a formal business-style summary.	Rewrote sections to better match the MyMart scenario and assignment requirements.
8	Claude Sonnet	Explain AUC-ROC score interpretation.	Provided a general explanation of ranking performance.	Accepted explanation but added dataset-specific context about the 3% fraud rate.

D2. Critical Reflection

AI tools played an important role in helping me complete this assessment, particularly when explaining machine learning concepts, reviewing code, and improving the clarity of my report. However, I found that the outputs were not always directly applicable to the MyMart dataset and often required careful checking before being used.

One example occurred when I asked an AI tool to evaluate the model's performance. The response highlighted the model's 97% accuracy and initially described this as a strong result. While the stdata set was technically correct, it was not appropriate for the MyMart problem because the dataset was highly imbalanced, with only 3.0% of transactions being chargebacks. From the course material, I understood that accuracy can be misleading in imbalanced classification tasks. A model that predicts every transaction as legitimate would still achieve roughly 97% accuracy while failing to detect any fraud. After reviewing the confusion matrix and class distribution, I decided to focus on recall, precision, and AUC-ROC instead of accuracy when evaluating the model.

Another issue occurred when Claude suggested that improving precision should be the primary objective because of the high number of false positives. Although this would reduce the number of legitimate customers being flagged, it conflicted with MyMart's business objective of minimising chargeback losses. Since each missed chargeback costs RM 85, reducing recalls too aggressively could increase financial losses. I therefore prioritised a high-recall strategy and discussed the trade-off between fraud detection and operational workload in the final report.

I also found that Gemini occasionally provided generic recommendations that were not tailored to the assignment. For example, it suggested collecting additional customer demographic information to improve model performance. While this might be useful in some situations, such data was not available in the MyMart dataset and introducing new variables was outside the scope of the assessment. Instead, I focused on the existing features provided in the dataset.

These experiences showed me that AI tools are valuable for generating ideas, checking code, and explaining concepts, but they should not be treated as unquestionable sources of information. I found it necessary to verify outputs against the dataset, business requirements, and machine learning principles covered in the course. The most useful approach was to treat AI as a starting point for analysis rather than a substitute for critical thinking and independent judgement.